Jay V. Kalinani

Postdoctoral Researcher | University of Illinois Urbana-Champaign

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Research profile. Computational astrophysicist using numerical relativity, general relativistic magnetohydrodynamics, and radiation transport to study compact object mergers, relativistic jets, and multimessenger signals. I also develop GPU-accelerated codes to perform large scale simulations of these systems.

Academic Appointments

Aug 2025–present **Postdoctoral Researcher**, University of Illinois Urbana-Champaign, USA
Advisor: Prof. Helvi Witek

Jan 2023–Jul 2025 **Postdoctoral Researcher**, Center for Computational Relativity and Gravitation, Rochester Institute of Technology, USA
Advisor: Prof. Manuela Campanelli

Jan–Dec 2022 **Postdoctoral Researcher**, INAF–Astronomical Observatory of Padova, Italy
Advisor: Dr. Riccardo Ciolfi

Education

2018–2021 **Ph.D. in Physics**, University of Padova, Italy, *summa cum laude*
Thesis: *Binary Neutron Star Mergers in the Era of Multi-Messenger Astrophysics*
Advisors: Dr. Riccardo Ciolfi and Prof. Jean-Pierre Zendri

2016–2018 **M.S. in Astrophysics**, AstroMundus Programme, 110/110 cum laude
Thesis: *GRMHD Simulations of Binary Neutron Star Mergers*

2014–2016 **M.S. in Applied Solid State Physics**, M. S. Ramaiah University of Applied Sciences, India, CGPA 8.17/10
Thesis: *Acoustic Scattering by Different Geometrical Structures*

2010–2014 **B.Tech. in Nuclear Science & Engineering**, SRM University, India, CGPA 9.01/10
Thesis: *Design Aspects of a Sodium Heated Steam Generator*

Selected Research Contributions

- **Code for exascale relativistic astrophysics.** Led development of `AsterX`, an open-source GPU-accelerated GRMHD code for dynamical spacetimes. Currently, leading complementary GPU-accelerated code development for neutrino transport, multiple black hole initial data, apparent horizon finding, passive particle tracking, and more. *(Kalinani et al., 2025)*
- **Relativistic outflows from neutron-star mergers.** Led simulations following two sequential polar outflows within one merger: first a dense massive-neutron-star-driven outflow and later a black-hole–disk jet compatible with a short gamma-ray burst. The simulations substantially reduced artificial-atmosphere effects and resolved the interaction between the outflows. Earlier work connected magnetically driven remnant winds to the blue kilonova of GW170817. *(Ciolfi & Kalinani, 2020; Kalinani et al., 2026)*
- **Microphysics and nucleosynthesis.** Contributed to merger simulations incorporating temperature-dependent nuclear equations of state, neutrino transport, and neutron-star spin, quantifying their effects on magnetic-field amplification, polar outflows, ejecta composition, and kilonova signatures. Co-mentored A. Wen. *(Nouri et al., 2026; Wen et al., 2026)*
- **Supermassive black-hole binaries.** Led simulations of a precessing, small mass ratio binary that connected disk crossings, jet precession, and intermittent jet weakening to quasi-periodic electromagnetic signatures; also contributed to inspiral through-merger accretion simulations predicting a transient luminosity enhancement. *(Ennoggi et al., 2025; Kalinani et al., 2026)*
- **Jet propagation across scales.** Contributed to cross-code handoffs from GRMHD merger remnants to large-scale relativistic jet simulations, linking central-engine and ejecta conditions to jet breakout and observable signatures. *(Pavan et al., 2021, 2023, 2025)*

Research Software and Infrastructure

Lead developer of GPU-accelerated research software for numerical relativity, GRMHD, neutrino transport, and particle-based tracking, primarily within the Einstein Toolkit ecosystem.

Lead developer and maintainer

2022–present **AsterX** – GPU-accelerated GRMHD for dynamical spacetimes; demonstrated scaling to 4,096 Frontier nodes.

2024–present **nuX** – GPU-accelerated moment-based neutrino radiation transport.

2026–present **BHClusterX** – Initial-data tools for black-hole cluster simulations in the Einstein Toolkit.

2026–present **FlowTracerX** – GPU-accelerated passive fluid-particle tracing for AsterX.
2026–present **ParticleAHFinderX** – GPU apparent-horizon finding with AMReX particles.
2025–present **AsterAI** – AI-assisted code-review tools for numerical relativity.

Developer and contributor

2020–present **Spritz** – GRMHD with neutrino leakage in fully dynamical spacetimes; developer.
2022–present **SpacetimeX** – Modules for spacetime initialization and evolution; developer.
2022–present **CarpetX** – AMReX-based Cactus driver for the Einstein Toolkit; contributor.
2021 **RePrimAnd** – Primitive-variable recovery scheme for GRMHD; contributor.

Research Funding and Computing Allocations

Research funding

2023–2025 **Co-Investigator**, JetOut: "Breakout, propagation, and observational signatures of short gamma-ray burst jets from binary neutron star mergers." Italy–USA scientific-cooperation project funded by the Italian Ministry of Foreign Affairs and International Cooperation; PI: Riccardo Ciolfi. **Total award: €131,000**.
Apr–Jul 2016 **CEFIPRA Research Grant**, Aquitaine–Karnataka pre-Ph.D. program. **Award: €4,500**.

High-performance computing

PI or Co-I on 29 allocations totaling more than 50.4 million CPU-hours and 1 million GPU-hours. **PI on 11 allocations** totaling approximately 15.5 million CPU-hours and 512,000 GPU-hours; PI awards are listed below.

Oct 2026–Oct 2027 **ACCESS Discover, DeltaAI** – 11,000 GPU-hours. *Evolution of a 50 black-hole cluster.*
Aug 2026–Feb 2027 **ALCF Director's Discretionary, Polaris** – 20,000 GPU-hours. *Scaling and Validation of Black-Hole Cluster Simulations.*
Mar 2026–Mar 2027 **ACCESS Discover, DeltaAI** – 11,000 GPU-hours. *GPU validation and benchmarking of AsterX/CarpetX with CoCalIDX initial data.*
Oct 2025–Oct 2026 **OLCF Director's Discretionary, Frontier** – 160,000 GPU-hours. *Benchmarking AsterX and CanudaX with magnetized neutron-star mergers.*
Feb 2025–Feb 2026 **TACC Pathways, Frontera** – 12.6 million CPU-hours. *Relativistic jet formation and bulk viscosity in magnetized neutron-star mergers.*
May 2023–May 2025 **OLCF Director's Discretionary, Frontier** – 320,000 GPU-hours. *Benchmarking the GPU-accelerated GRMHD code AsterX.*
Dec 2022–Jun 2023 **INAF–CINECA Class B, Galileo 100** – approximately 375,000 CPU-hours. *Binary neutron-star mergers with Spritz and the APR4 equation of state.*
Jul–Dec 2022 **EuroHPC Benchmark Access, Discoverer** – approximately 1 million CPU-hours. *Binary neutron-star merger simulations with Spritz.*
Mar–Dec 2022 **CINECA Class C, Galileo 100** – approximately 100,000 CPU-hours. *Magnetized binary neutron-star merger simulations with Spritz.*
Aug 2021–May 2022 **CINECA Class C, Galileo 100** – approximately 192,000 CPU-hours. *Fishbone–Moncrief disk simulations with Spritz.*
Apr 2020–Apr 2021 **CINECA Class B, Marconi 100 and Galileo** – approximately 1.25 million CPU-hours. *Binary neutron-star merger simulations with Spritz.*

Publications

18 papers: 16 published, and 2 submitted; 731 citations; h-index 13. Citation metrics from NASA ADS, October 2026.

Submitted manuscripts

2026 **Jet variability in small mass ratio supermassive black hole binaries**.
J. V. Kalinani, M. Campanelli, M. Chabanov, M. C. de Simone, C. O. Lousto, H. Witek, and Y. Zlochower.
Submitted to ApJL. arXiv:2609.24837.
2026 **Binary Neutron Star Merger Simulations with Microphysical Equation of State using Spritz**.
F. H. Nouri, **J. V. Kalinani**, B. Giacomazzo, R. Ciolfi, and A. Perego.
Submitted to CQG. arXiv:2608.23310.

Published peer-reviewed articles

2026 **Magnetic eruption and nucleosynthesis in GRνMHD simulations of spinning neutron star mergers**.
A. Wen, **J. V. Kalinani**, M. Chabanov, M. Campanelli, R. Ciolfi, and Y. Zlochower.
PRD 114, 083023.
2026 **Jet-environment interaction after delayed collapse in binary neutron star mergers**.
J. V. Kalinani, R. Ciolfi, M. Campanelli, B. Giacomazzo, A. Pavan, A. Wen, and Y. Zlochower.
ApJL 1000, L35.
2026 **Accelerating numerical relativity simulations with new multistep fourth-order Runge–Kutta methods**.
L. T. Sanches, S. R. Brandt, **J. V. Kalinani**, L. Ji, and E. Schnetter.
CQG 43, 065010.

Published peer-reviewed articles (continued)

2025 **Relativistic gas accretion onto supermassive black hole binaries from inspiral through merger**.
L. Ennoggi, M. Campanelli, Y. Zlochower, S. C. Noble, J. Krolik, F. Cattorini, **J. V. Kalinani**, V. Mewes, M. Chabanov, L. Ji, and M. C. de Simone.
PRD 112, 063009.

2025 **Role of injection parameters in jet propagation through realistic binary neutron star merger environments**.
A. Pavan, R. Ciolfi, E. Dreas, and **J. V. Kalinani**.
MNRAS 540, 1345–1362.

2025 **AsterX: a new open-source GPU-accelerated GRMHD code for dynamical spacetimes**.
J. V. Kalinani, L. Ji, L. Ennoggi, F. G. Lopez Armengol, L. T. Sanches, B. J. Tsao, S. R. Brandt, M. Campanelli, R. Ciolfi, B. Giacomazzo, R. Haas, E. Schnetter, and Y. Zlochower.
CQG 42, 025016.

2023 **Jet-environment interplay in magnetized binary neutron star mergers**.
A. Pavan, R. Ciolfi, **J. V. Kalinani**, and A. Mignone.
MNRAS 524, 260–275.

2022 **Implementing a new recovery scheme for primitive variables in the GRMHD code Spritz**.
J. V. Kalinani, R. Ciolfi, W. Kastaun, B. Giacomazzo, F. Cipolletta, and L. Ennoggi.
PRD 105, 103031.

2022 **Turbulent magnetic field amplification in binary neutron star mergers**.
C. Palenzuela, R. Aguilera-Miret, F. Carrasco, R. Ciolfi, **J. V. Kalinani**, W. Kastaun, B. Miñano, and D. Viganò.
PRD 106, 023013.

2021 **Short gamma-ray burst jet propagation in binary neutron star merger environments**.
A. Pavan, R. Ciolfi, **J. V. Kalinani**, and A. Mignone.
MNRAS 506, 3483–3498.

2021 **Spritz: General Relativistic Magnetohydrodynamics with Neutrinos**.
F. Cipolletta, **J. V. Kalinani**, E. Giangrandi, B. Giacomazzo, R. Ciolfi, L. Sala, and B. Giudici.
CQG 38, 085021.

2021 **Robust Recovery of Primitive Variables in Relativistic Ideal Magnetohydrodynamics**.
W. Kastaun, **J. V. Kalinani**, and R. Ciolfi.
PRD 103, 023018.

2020 **Magnetically driven baryon winds from binary neutron star merger remnants & blue kilonova of August 2017**.
R. Ciolfi and **J. V. Kalinani**.
ApJL 900, L35.

2020 **Spritz: a new fully general-relativistic magnetohydrodynamic code**.
F. Cipolletta, **J. V. Kalinani**, B. Giacomazzo, and R. Ciolfi.
CQG 37, 135010.

2019 **First 100 ms of a long-lived magnetized neutron star formed in a binary neutron star merger**.
R. Ciolfi, W. Kastaun, **J. V. Kalinani**, and B. Giacomazzo.
PRD 100, 023005.

2014 **Thermal Analysis of a Mobile Hot Cell Cask**.
J. V. Kalinani and A. R. Kumar.
Int. J. Innov. Res. Sci. Eng. Technol. 3(11), 17575–17584.

Honors and Awards

2018–2021 **CARIPARO Ph.D. Scholarship**, Cariparo Foundation, Italy (€99,800).

Apr 2019 **Best Ph.D. Student Talk**, PHAROS Conference, Playa de Aro, Spain.

2016–2018 **AstroMundus Consortium Scholarship** (€43,400).

Oct 2016 **Best Master's Thesis Award**, M.S. Ramaiah University, India.

2010–2012 **Department Scholarship for Rank 1**, SRM University, India.

2007–2008 **AFS Youth Exchange and Study Scholarship**, one-year high-school exchange in San Diego, USA.

Teaching Experience

UIUC **Guest lecturer, undergraduate Classical Mechanics** – lectures on drag forces and harmonic oscillators. Course instructors: Profs. Hector O. Silva and Helvi Witek.

RIT **Guest lecturer, graduate astrophysics** – lectures on general relativistic magnetohydrodynamics and numerical methods for conservation laws in *Astrophysical Flows* and *Fundamentals of Theoretical Astrophysics I*. Course instructor: Prof. Manuela Campanelli.

Einstein Toolkit **Workshop instructor** – designed and delivered hands-on tutorials on CarpetX and AsterX at community workshops and schools.

Research Mentoring

Graduate students **Co-mentor**: Allen Wen (RIT, Dec 2023–present); Amara Nusbickel (RIT, Mar 2025–Mar 2026); Maxwell Rizzo (UIUC, Aug 2025–present); Noora Ghadiri (UIUC, Mar 2026–present); Freddy Pardoe (UIUC, Mar 2026–present).

Hackathons **Mentor**: Bing-Jyun Tsao (2023) and Maxwell Rizzo (2025) during Einstein Toolkit hackathons.

Selected Professional Development

Jul–Aug 2026 **Argonne Training Program on Extreme-Scale Computing** – selected intensive program in exascale application design, implementation, and execution.

Sep 2025–Apr 2026 **Scholarship and Teaching for Engineering Postdocs Program** – faculty-preparation program in teaching, mentoring, and research.

Academic Service and Research Leadership

Organization Local and scientific organizer, North American Einstein Toolkit Workshop and School (UIUC, 2026); local organizer, CSSI Einstein Toolkit Hackathon (2025); organizer, CCRG Friday lunch seminars.

Leadership Team lead, AsterX/CarpetX group, Frontier Hackathon (2023).

Peer review Reviewer for the U.S. National Science Foundation, Physical Review journals, *Classical and Quantum Gravity*, and *Astronomy & Astrophysics*.

Memberships American Physical Society (DGRAV and DCOMP); LISA Consortium; Theory and Computational Astrophysics Network; Indian Association for General Relativity and Gravitation.

Presentations

46 presentations: 23 invited talks, seminars, and tutorials; 23 contributed talks and posters.

Selected invited presentations

Jul 2026 **Relativistic outflows from compact-object binaries**. New Frontiers in Strong Gravity, Benasque, Spain, online.

Mar 2026 **General relativistic magnetohydrodynamics on GPUs**. APS Global Physics Summit, Denver, USA.

Feb 2026 **Towards end-to-end modelling of jets from BNS mergers**. Nuclear Physics from Multi-Messenger Mergers Seminar, online.

Apr 2025 **End-to-end modelling of jets from BNS mergers**. The 7th Neighbourhood Workshop, Penn State University, USA. [plenary]

Oct 2024 **End-to-end modelling of jets from BNS mergers**. 130th Topical Symposium, New York State Section of the APS, SUNY Polytechnic Institute, USA.

Jul 2024 **AsterX: a new open-source GPU-accelerated GRMHD code for dynamical spacetimes**. Solving the Boltzmann Equation for Neutrino Transport, ICERM, USA.

Aug 2023 **Binary Neutron Star Merger Simulations: Long-Lived Remnants, Short Gamma-Ray Bursts & Kilonovae**. INT 23-2 Workshop, University of Washington, USA.

May 2023 **Binary Neutron Star Merger Simulations: Long-Lived Remnants, Short Gamma-Ray Bursts & Kilonovae**. Numerical Relativity Seminar, online.

Additional invited talks, seminars, and tutorials

Dec 2025 **Tutorial on the AsterX code**. European Einstein Toolkit Meeting, online. [tutorial]

Oct 2025 **End-to-end modelling of jets from BNS mergers**. Astrophysics, Relativity, and Cosmology Seminar, UIUC, USA.

Aug 2025 **nuX-M1: a new GPU-accelerated moment-based neutrino radiation transport code**. Einstein Toolkit CSSI Workshop, UIUC, USA.

Jun 2025 **The AsterX and nuX code**. North American Einstein Toolkit Workshop, University of Texas at Austin, USA.

Oct 2024 **End-to-end modelling of jets from BNS mergers**. NEOSGrav Conference, Goa, India.

Jul 2024 **AsterX: a new open-source GPU-accelerated GRMHD code for dynamical spacetimes**. North American Einstein Toolkit Workshop, Louisiana State University, USA.

Jul 2023 **Introduction to GRMHD and the AsterX code**. North American Einstein Toolkit School, RIT, USA.

Jul 2023 **The CarpetX/AsterX code**. North American Einstein Toolkit School, RIT, USA. [tutorial]

Oct 2022 **Magnetized BNS merger simulations and prospects of launching SGRB jets**. EMMI–IReNA GSI Workshop, online.

Jun 2022 **Binary Neutron Star Merger Simulations with WhiskyMHD and Spritz**. TCAN Meeting, online.

Jun 2022 **The CarpetX Framework**. North American Einstein Toolkit Workshop, online. [tutorial]

Jun 2022 **The Spritz Code: GRMHD with Neutrinos**. North American Einstein Toolkit Workshop, online.

Jul 2021 **Magnetized BNS Merger Simulations with Long Post-Merger Evolution**. TCAN BNS/BH–NS Merger Workshop.

Jul 2021 **Binary Neutron Star Merger Simulations: Long-Lived Remnant, Short Gamma-Ray Bursts, and Kilonovae**. 16th Marcel Grossmann Meeting, online.

Jun 2021 **GRMHD Simulations of Binary Neutron Star Mergers**. ECT* Workshop: Neutron Stars as Multi-Messenger Laboratories for Dense Matter.

Contributed talks and posters

Jul 2026 **Jet variability in small mass ratio supermassive black-hole binaries**. UK & Ireland Discs Meeting, online. [poster]

Contributed talks and posters (continued)

Jul 2026 **Jet variability in small mass ratio supermassive black-hole binaries**. 16th International LISA Symposium, College Park, USA.
Mar 2026 **Jet-environment interaction after delayed collapse in BNS mergers**. 3rd International Online Conference on Universe, online.
Dec 2024 **AsterX: a new open-source GPU-accelerated GRMHD code for dynamical spacetimes**. Numerical Relativity Community, online.
Apr 2024 **End-to-end modelling of jets from binary neutron star mergers**. APS April Meeting, Sacramento, USA.
Apr 2023 **AsterX: a new open-source GPU-accelerated GRMHD code for dynamical spacetimes**. APS April Meeting, Minneapolis, USA.
Jul 2022 **Magnetized BNS merger simulations and prospects of launching SGRB jets**. COSPAR 44th Scientific Assembly, Athens, Greece.
May 2022 **Binary Neutron Star Merger Simulations with GRMHD code Spritz**. PHAROS Conference, Rome, Italy.
Apr 2022 **Binary Neutron Star Merger Simulations with GRMHD code Spritz**. APS April Meeting, online.
Nov 2021 **Magnetars born in BNS mergers, Short GRBs and Kilonovae**. IAU Symposium 363, online.
Jul 2021 **A new primitive variable recovery scheme in GRMHD code Spritz**. North American Einstein Toolkit Summer School.
Apr 2021 **The Spritz Code: Improving the Robustness of GRMHD**. APS April Meeting, online.
Feb 2021 **A new primitive variable recovery scheme in GRMHD code Spritz**. PHAROS–CompOSE Conference, online.
Feb 2021 **Binary Neutron Star Merger Simulations: Long-Lived Remnant, Short Gamma-Ray Bursts and Kilonovae**. TEONGRAV Workshop, online.
Dec 2020 **Role of Magnetic Fields in Binary Neutron Star Merger Simulations**. IAGRG Meeting, online.
Jul 2020 **Role of Magnetic Fields in Binary Neutron Star Merger Simulations**. TCAN Binary Neutron Stars Workshop.
Apr 2020 **GRMHD Simulations of BNS Mergers: 100 ms of Long-Lived Remnant Evolution**. APS April Meeting, online.
Sep 2019 **GRMHD Simulations of BNS Mergers: 100 ms of Long-Lived Remnant Evolution**. European Einstein Toolkit Meeting, London, UK.
Apr 2019 **GRMHD Simulations of BNS Mergers: 100 ms of Long-Lived Remnant Evolution**. PHAROS Conference, Playa de Aro, Spain.
Mar 2019 **GRMHD Simulations of BNS Mergers**. PHAROS Ph.D. School, University of Jena, Germany. [poster]
Jan 2019 **GRMHD Simulations of BNS Mergers**. IAGRG Meeting, BITS Pilani Hyderabad, India. [poster]
Jun 2018 **GRMHD Simulations of BNS Mergers**. AstroMundus Retreat, Obergurgl, Austria. [poster]
Aug 2015 **Precision Particle Tracking**. Center for Axion and Precision Physics, KAIST, South Korea. [poster]

Public Engagement

2020–present **Volunteer,** Skype a Scientist – interactive astronomy sessions with classrooms and families.
Apr 2023 **Exhibitor, Imagine RIT: Creativity and Innovation Festival** – represented the astrophysics department.
May 2020 **Organizer and speaker,** Let's Talk Astronomy – online public talk, *Galaxies, Stars and Planets*.
Apr 2020 **Contributor, Astrobites** – *Short Gamma Ray Burst Central Engines: A Curious Case of Time-Reversal*.

Additional Community Service

2023–2025 Dog-care volunteer, Rochester Animal Services, Rochester, USA.
2020–2025 Active listener, 7 Cups.
2014–2016 Education-support volunteer, Make A Difference, Bangalore, India.
2012–2014 Head of Operations and Co-Director, Project Book-Wings, The Green Nest, Chennai, India.
2012–2013 Volunteer, Child Rights and You, Bangalore and Chennai, India.

Academic References

Current advisor **Prof. Helvi Witek**, University of Illinois Urbana-Champaign
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Former advisor **Prof. Manuela Campanelli**, Center for Computational Relativity and Gravitation, Rochester Institute of Technology
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Ph.D./postdoc advisor **Dr. Riccardo Ciolfi**, INAF–Astronomical Observatory of Padova and INFN Padova
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